LESSON 6.8

Generate Equivalent Expressions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use the commutative, associative, and identity properties to rewrite expressions.

WHY THIS MATTERS You're a band director arranging musicians on stage. You have 3 guitarists, 5 drummers, and 2 keyboard players. Whether you seat the guitarists first or the drummers first, you still have 3 + 5 + 2 = 10 musicians. The order you arrange them doesn't change the total!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Properties of Operations *Spanish: Propiedades de las operaciones*	Rules that let you reorder or regroup numbers without changing an expression's value.	3 + 5 = 5 + 3 (commutative property).
Commutative Property *Spanish: Propiedad conmutativa*	You can change the order and get the same answer.	a + b = b + a
Associative Property *Spanish: Propiedad asociativa*	You can change the grouping and get the same answer.	(a + b) + c = a + (b + c)
Identity Property *Spanish: Propiedad de identidad*	Adding 0 or multiplying by 1 keeps the same value.	a + 0 = a · a × 1 = a
Property *Spanish: Propiedad*	A rule that is always true in math.	Commutative works for + and ×, but NOT for − or ÷ (since 5 − 3 = 2 but 3 − 5 = −2)
Distributive Property *Spanish: Propiedad distributiva*	Multiplying a number by everything inside the parentheses: a(b + c) = ab + ac.	a(b + c) = ab + ac

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Simplify 4(m + 2) + 3m.

1. Distribute the 4: 4m + 8 + 3m.
2. Move like terms together: 4m + 3m + 8.
3. Combine: 7m + 8. Test m = 1: 4(3) + 3 = 15 and 7 + 8 = 15.

Answer: 7m + 8

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Look at 6 + 9 and 9 + 6. Do they give the same total?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I used the _____ Property so that I could _____ .
- After distributing, the expression became _____ .
- The like terms _____ and _____ combine to make _____ .
- My rewritten expression is equivalent because _____ .

Word bank Commutative | Associative | Identity | Distributive | property | equivalent | reorder | regroup | distribute | combine | like terms | value

WATCH OUT!

A common mistake

A common mistake in Properties of Operations is applying the commutative property where it does not belong, or confusing it with the associative property. Students often assume 15 − 8 = 8 − 15 (it does not: 7 ≠ −7

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Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each equation by the property it shows.

Groups: Commutative | Associative | Identity

_____ 9 + 4 = 4 + 9

_____ 7 × 1 = 7

_____ (5 + 1) + 3 = 5 + (1 + 3)

HINT For each equation ask: did the order swap, did the parentheses move, or did a 0 or 1 leave the number unchanged?

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which property is shown? 5 + 13 = 13 + 5

(A) Commutative Property
(B) Associative Property
(C) Identity Property
(D) Distributive Property

HINT Compare the two sides: same numbers, same operation. What changed between left and right?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which property is shown? $(6 \times 3) \times 2 = 6 \times (3 \times 2)$

(A) Commutative Property
(B) Identity Property
(C) Associative Property
(D) Distributive Property

HINT Read both sides carefully: 6, 3, 2 stay in the same order. Only the parentheses moved.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which property is shown? $47 + 0 = 47$

(A) Commutative Property
(B) Associative Property
(C) Distributive Property
(D) Identity Property

HINT Notice what was added: 0. Did adding it change 47 at all?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which property lets you rearrange 8×5 to 5×8?

(A) Associative Property
(B) Identity Property
(C) Distributive Property
(D) Commutative Property

HINT The factors 8 and 5 swapped positions. Nothing else changed — no parentheses, no new numbers.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Statement** 4 + 0 = 4
2. **Student's label** This shows the Commutative Property.
3. **Reasoning** The numbers 4 and 0 switched places, so it must be commutative.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** What does adding 0 to a number do to its value?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How can you tell the difference between the commutative and associative properties? What clue helps you decide?

SENTENCE STARTER The ___ property changes the ___, while the ___ property changes the ___. I can tell by looking at whether the ___ moved or the ___ changed.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.8

Generate Equivalent Expressions

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Match each equation with the property it demonstrates.

Groups: Commutative | Associative | Identity

_____ a + b = b + a

_____ x × 1 = x

_____ (m + n) + p = m + (n + p)

_____ c × d = d × c

_____ y + 0 = y

_____ (a × b) × c = a × (b × c)

2 **Write the letter of the matching item on each line.**

Match each equation to its property.

_____ 7 + 9 = 9 + 7

_____ (2 × 5) × 3 = 2 × (5 × 3)

_____ 15 × 1 = 15

_____ 0 + 23 = 23

_____ 6 × 4 = 4 × 6

_____ (8 + 3) + 7 = 8 + (3 + 7)

(A) Commutative (Add)

(B) Associative (Mult)

(C) Identity (Mult)

(D) Identity (Add)

(E) Commutative (Mult)

(F) Associative (Add)

3 **Complete the table. Show how you found each value.**

Use properties to rewrite each expression in a way that makes mental math easier. Then calculate.

Original	Rearranged (using properties)	Value
25 × 7 × 4	25 × 4 × 7 = 100 × 7	
19 + 36 + 1	19 + 1 + 36 = 20 + 36	
50 × 9 × 2	50 × 2 × 9 = 100 × 9	

SHOW YOUR WORK

4 **Decide whether each pair balances. Check Yes or No, then explain.**

Associative Property of Addition: same value (13), different grouping

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Statement** (2 + 3) + 4 = (3 + 2) + 4
2. **Student's label** This shows the Associative Property.
3. **Reasoning** There are parentheses, so the grouping must have changed.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How can you tell the difference between the commutative and associative properties? What clue helps you decide?

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LESSON 6.8

Generate Equivalent Expressions

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

For each expression, use the commutative and/or associative properties to rearrange and simplify the mental math. Show which property you used.

Expression	Rearranged	Property Used	Value
2 × 13 × 50			
45 + 27 + 55			

SHOW YOUR WORK

2 **Complete the table. Show how you found each value.**

6.AT.7 asks you to use properties to GENERATE an equivalent expression, not just to name a property — try it with variables instead of just numbers. For each expression, apply the property named to write an equivalent expression.

Expression	Property to apply	Equivalent expression
y + y + y	Repeated addition → multiplication	
(x + 4) + 6	Associative Property	
5 + n	Commutative Property	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

3 Find the mistake. Explain it, then show the correct work.

Find the Property Error

1. **Statement** 15 − 8 = 8 − 15
2. **Student's claim** This shows the Commutative Property of Subtraction.
3. **Reasoning** I switched the order, so it must be commutative.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

4 Answer in complete sentences.

Explain why the commutative property works for addition and multiplication but NOT for subtraction and division. Give a specific number example for each operation to prove your point.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How can you tell the difference between the commutative and associative properties? What clue helps you decide?

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